Sewage Treatment Methods

Sewage treatment defined as an artificial process in which sewage is subjected to remove / alter its constituents to render it less offensive. There are three levels of wastewater treatment (Figure 9.7): primary, secondary, and tertiary (or advanced).

1. Primary treatment
2. Secondary treatment
3. Tertiary / Final treatment

Primary treatment

Primary treatment removes about 60 percent of total suspended solids and about 35 percent of BOD; dissolved impurities are not removed. It is the physical method which remove large floating and suspended solids from sewage water. Example: papers, leaves, bottles, rocks, pieces of metal or wood. These objects are removed by passing the sewage through screens (Figure 9.8) and grit chambers. The screened water is then sent to settling tanks or basin, where the suspended solids are allowed to settle as primary sludge. Materials such as oil or grease, which float on the surface, are removed with a skimmer. The liquid wastewater remaining in the settling tank or basin is then ready for secondary treatment. The fluid from primary treatment is called primary effluent.



Figure 9.8: Bar Screening in Sewage Treatment

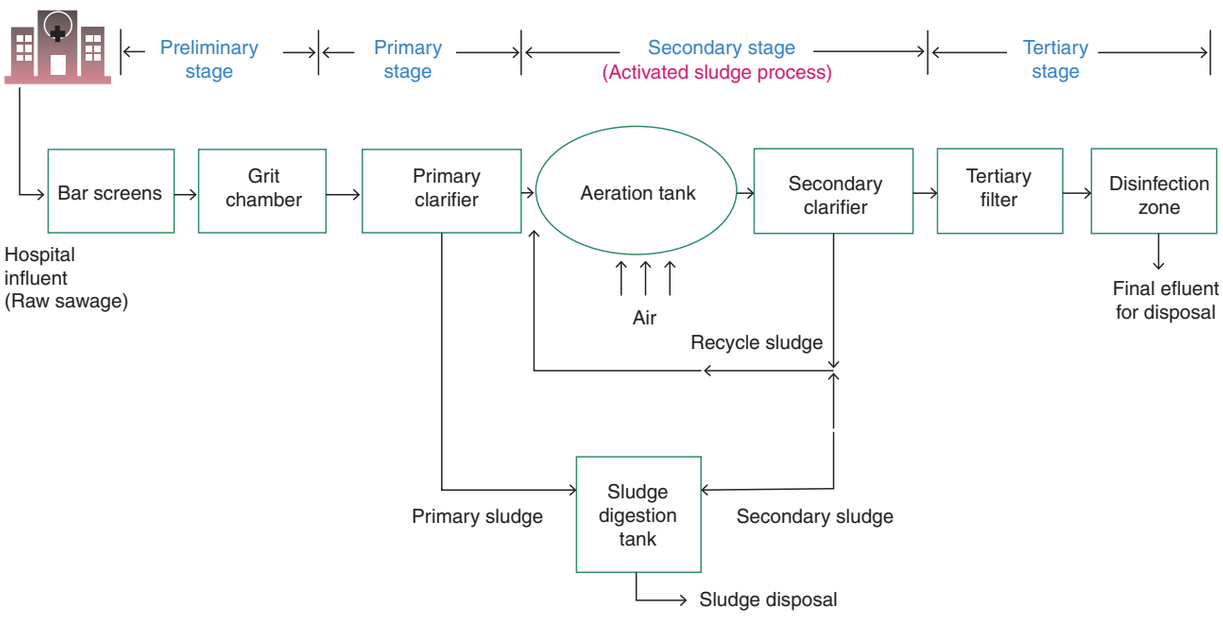


Figure 9.7: Schematic diagram of waste water treatment

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